A Level Mathematics 9709 · M1 Mechanics

全套中英文详细学习手册

Detailed bilingual student manual · concept, derivation, worked examples and exam method

本册共 8 个独立课时。每课按三页展开：第一页建立概念、推导方法并完成基础例题；第二页处理 Cambridge 风格应用、英文表达、评分点与易错辨析；第三页扩展考纲边界、变式推演与迁移任务。题目均为依据考纲原创的训练示例，不复制历年试题。

使用范围与版本 | Scope

0580 册依据 Cambridge IGCSE Mathematics 0580（20252027）的 Core/Extended 分层；9709 册依据 Cambridge International AS & A Level Mathematics 9709（20262027）。M2
单独标为拓展/旧体系衔接材料，不作为现行独立考试组件。

目录丨Contents

1. 运动学：位移速度加速度丨Kinematics

2. 速度时间图丨Velocity–time graphs

3. 力与牛顿定律丨Forces and Newton’s laws

4. 斜面与摩擦丨Inclined planes and friction

5. 连接粒子与滑轮丨Connected particles

6. 动量与冲量丨Momentum and impulse

7. 功、能、功率丨Work, energy and power

8. M1 综合建模丨Mechanics synthesis

第 1 课 | 运动学：位移速度加速度 | Kinematics

Lesson 1 of 8 · Detailed bilingual learning unit

概念拆解 | Concept and meaning

速度是位移变化率，加速度是速度变化率；向量符号方向必须一致。

English core explanation. Choose and state a positive direction, isolate the body or system, and write equations with signs that match the diagram. Formulae such as SUVAT, F=ma and conservation laws are valid only when their modelling assumptions are satisfied.

为什么方法成立 | Reasoning behind the method

先选正方向，再列s,u,v,a,t；恒加速度才使用SUVAT。

把知识点拆成可执行步骤 | A reliable method

1. Read the command word and identify the target quantity. 先圈出 find, show, prove, sketch, hence 或 explain。
2. Translate the information into equations, a labelled diagram, a table or a distribution.
3. Choose a theorem or formula only after checking its assumptions.
4. Keep exact values through intermediate steps; round once at the end.
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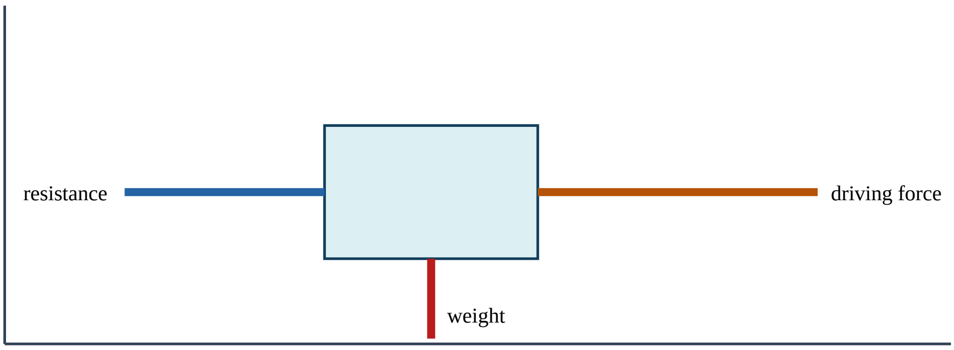

例题一：从题意到完整解答 | Worked example 1

物体u=5 m/s，a=3 m/s，4秒后速度与位移。

v=5+3 × 4=17；s=5 × 4+0.5 × 3 × 16=44 m。

Reading the mathematics. The first line identifies the structure; the middle lines show the transformation or calculation; the final line answers the question in context. Do not hide essential reasoning inside the calculator.

第 1 课（续） | 运动学：位移速度加速度 | Kinematics

例题二：Cambridge 风格应用 | Cambridge-style application

A particle slows from 20 to 8 m/s in 6 s. Find acceleration.

a=(820)/6=2 m/s.

评分点如何产生 | Where marks come from

多数结构题把分数分给不同层次：建立正确模型或公式通常得到 method mark；正确代入并推进得到后续 method/accuracy marks；准确答案、单位、范围或结论得到 final accuracy mark。若前一步算错但方法仍一致，后续常可获得 follow-through，因此必须写出过程。证明题不能只写计算器结果，show that 题还要到达题目指定的精确形式。

English mathematical communication

Use complete mathematical sentences: "Substituting ... gives ...", "Therefore ...", "Since the discriminant is zero ...", "The events are independent because ...", or "The negative sign shows motion opposite to the chosen positive direction." When a question says hence, use the previous result. When it says exact, do not replace π, e, logarithms or radicals by rounded decimals.

易错辨析 | Common errors and repairs

条件未检查：把只适用于直角三角形、恒加速度、独立事件或正态模型的公式随意套用。修正：公式前写出适用结构。

中途过早取整：后续答案超出允许误差。修正：保留精确值或计算器完整储存值。

只给答案：丢失方法分。修正：至少展示关键公式、代入与一次等价变形。

符号或单位遗漏：尤其是向量方向、平方/立方单位、概率区间和角度模式。

漏解或增根：解三角、模、平方与对数方程后，回到原定义域逐一检查。

再推一步 | Extension and connection

把本课与相邻知识连接：同一对象通常可从符号、图像、数值表和现实情境四种表示来理解。若代数式得到的符号与图像趋势不一致，应回查运算；若概率或测量答案超出合理范围，应先检查模型而不是盲信计算器。这样的交叉检查是从“会算”走向“会解题”的关键。

> Exam-ready summary. Define or identify → model → calculate with visible steps → interpret → check. Your final statement should answer the exact question asked, with appropriate accuracy and units.

知识边界与相邻考点 | Scope and connections

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> Detailed English explanation. Choose and state a positive direction, isolate the body or system, and write equations with signs that match the diagram. Formulae such as SUVAT, F=ma and conservation laws are valid only when their modelling assumptions are satisfied.

变式推演 | Variation and transfer

若把例题中的具体数换成字母，应先建立一般关系再代值；若改变范围、方向、图形性质或随机条件，则必须重新检查原方法的适用条件。把答案遮住，尝试只凭题干写出第一行模型；第一行正确，后续计算才有方向。

第三道示范：审题、计算、复核 | Worked reasoning example

重新审视本课基础题：物体u=5 m/s，a=3 m/s，4秒后速度与位移。

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Cambridge 题型如何包装 | How the idea is examined

选择题常把典型失误设计成干扰项：漏掉第二个解、混淆百分数基数、把面积比当长度比、把独立当互斥或遗漏连续性修正。结构题常先要求 show/find 一个中间结果，再用 hence
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英文作答骨架 | Useful response frames

“From the given information, ...” → “Using ..., we obtain ...” → “Substituting and simplifying gives ...” → “Therefore the required value is ...” . For interpretation, write “This means that ... in the context of the question.” For validation, write “The value lies in the stated domain and satisfies the original condition.”

第三个迁移任务 | Transfer task

A particle slows from 20 to 8 m/s in 6 s. Find acceleration.

独立完成后对照：a=(820)/6=2 m/s.

> Exam check. Have you answered the command word, shown the decisive line of working, kept suitable accuracy, included units or direction, and considered every valid solution?

第 2 课 | 速度时间图 | Velocity–time graphs

Lesson 2 of 8 · Detailed bilingual learning unit

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v–t图斜率是加速度，带符号面积是位移；路程需面积绝对值。

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为什么方法成立 | Reasoning behind the method

分段求三角形/梯形面积，穿过时间轴时分开。

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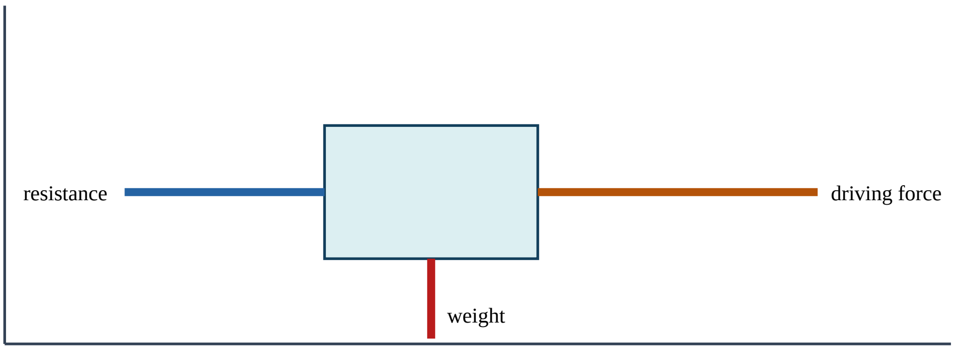

例题一：从题意到完整解答 | Worked example 1

速度从0线性增至12再在4秒降至0，求位移。

三角形面积=0.5 × 4 × 12=24 m。

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第 2 课（续）| 速度时间图 | Velocity–time graphs

例题二：Cambridge 风格应用 | Cambridge-style application

Explain difference between distance and displacement.

Distance totals all movement; displacement is signed change in position.

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中途过早取整：后续答案超出允许误差。修正：保留精确值或计算器完整储存值。
只给答案：丢失方法分。修正：至少展示关键公式、代入与一次等价变形。
符号或单位遗漏：尤其是向量方向、平方/立方单位、概率区间和角度模式。
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第三个迁移任务 | Transfer task

Explain difference between distance and displacement.

独立完成后对照：Distance totals all movement; displacement is signed change in position.

> Exam check. Have you answered the command word, shown the decisive line of working, kept suitable accuracy, included units or direction, and considered every valid solution?

第 3 课 | 力与牛顿定律 | Forces and Newton’ s laws

Lesson 3 of 8 · Detailed bilingual learning unit

概念拆解 | Concept and meaning

合力产生加速度：F=ma；作用反作用力作用在不同物体上。

English core explanation. Choose and state a positive direction, isolate the body or system, and write equations with signs that match the diagram. Formulae such as SUVAT, F=ma and conservation laws are valid only when their modelling assumptions are satisfied.

为什么方法成立 | Reasoning behind the method

隔离物体画free-body diagram，只画作用在该物体上的力。

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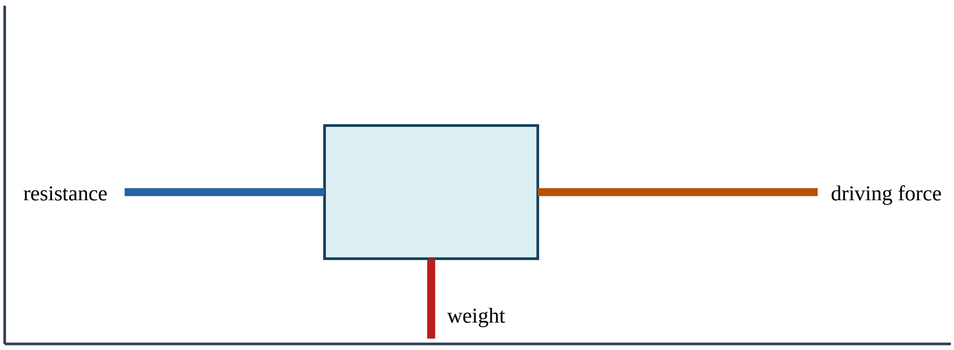

例题一：从题意到完整解答 | Worked example 1

质量4 kg受向右18 N、向左6 N，求加速度。

合力12 N向右，a=12/4=3 m/s向右。

Reading the mathematics. The first line identifies the structure; the middle lines show the transformation or calculation; the final line answers the question in context. Do not hide essential reasoning inside the calculator.

第 3 课（续）｜力与牛顿定律｜Forces and Newton’ s laws

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A 5 kg body is in equilibrium under horizontal forces. One is 12 N right. Find the other.

12 N left, because resultant force is zero.

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只给答案：丢失方法分。修正：至少展示关键公式、代入与一次等价变形。
符号或单位遗漏：尤其是向量方向、平方/立方单位、概率区间和角度模式。
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第三个迁移任务 | Transfer task

A 5 kg body is in equilibrium under horizontal forces. One is 12 N right. Find the other.

独立完成后对照：12 N left, because resultant force is zero.

> Exam check. Have you answered the command word, shown the decisive line of working, kept suitable accuracy, included units or direction, and considered every valid solution?

第 4 课丨斜面与摩擦丨Inclined planes and friction

Lesson 4 of 8 · Detailed bilingual learning unit

概念拆解丨Concept and meaning

重力沿斜面分量mgsinθ，法向分量mgcosθ；摩擦反对相对运动趋势。

English core explanation. Choose and state a positive direction, isolate the body or system, and write equations with signs that match the diagram. Formulae such as SUVAT, F=ma and conservation laws are valid only when their modelling assumptions are satisfied.

为什么方法成立丨Reasoning behind the method

先判断可能运动方向，再放摩擦；极限摩擦F=μR仅在limiting equilibrium。

把知识点拆成可执行步骤丨A reliable method

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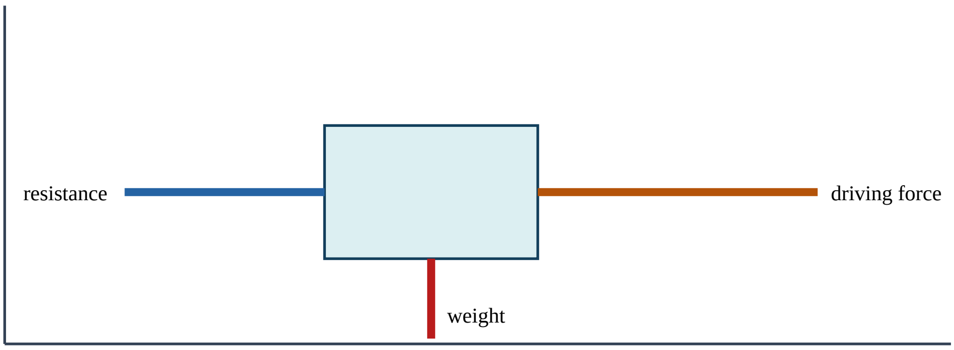

例题一：从题意到完整解答丨Worked example 1

2 kg块在30°光滑斜面上的加速度。

沿面力=2g sin30°，所以a=g sin30°=4.9 m/s。

Reading the mathematics. The first line identifies the structure; the middle lines show the transformation or calculation; the final line answers the question in context. Do not hide essential reasoning inside the calculator.

第 4 课（续）｜斜面与摩擦 | Inclined planes and friction

例题二：Cambridge 风格应用 | Cambridge-style application

Why is friction not always μR?

$F \leqslant \mu R$; equality holds only when motion is impending or limiting.

评分点如何产生 | Where marks come from

多数结构题把分数分给不同层次：建立正确模型或公式通常得到 method mark；正确代入并推进得到后续 method/accuracy marks；准确答案、单位、范围或结论得到 final accuracy mark。若前一步算错但方法仍一致，后续常可获得 follow-through，因此必须写出过程。证明题不能只写计算器结果，show that 题还要到达题目指定的精确形式。

English mathematical communication

Use complete mathematical sentences: “Substituting … gives …”, “Therefore …”, “Since the discriminant is zero …”, “The events are independent because …”, or “The negative sign shows motion opposite to the chosen positive direction.” When a question says hence, use the previous result. When it says exact, do not replace π, e, logarithms or radicals by rounded decimals.

易错辨析 | Common errors and repairs

条件未检查：把只适用于直角三角形、恒加速度、独立事件或正态模型的公式随意套用。修正：公式前写出适用结构。

中途过早取整：后续答案超出允许误差。修正：保留精确值或计算器完整储存值。

只给答案：丢失方法分。修正：至少展示关键公式、代入与一次等价变形。

符号或单位遗漏：尤其是向量方向、平方/立方单位、概率区间和角度模式。

漏解或增根：解三角、模、平方与对数方程后，回到原定义域逐一检查。

再推一步 | Extension and connection

把本课与相邻知识连接：同一对象通常可从符号、图像、数值表和现实情境四种表示来理解。若代数式得到的符号与图像趋势不一致，应回查运算；若概率或测量答案超出合理范围，应先检查模型而不是盲信计算器。这样的交叉检查是从“会算”走向“会解题”的关键。

> Exam-ready summary. Define or identify → model → calculate with visible steps → interpret → check. Your final statement should answer the exact question asked, with appropriate accuracy and units.

知识边界与相邻考点 | Scope and connections

重力沿斜面分量mgsin θ，法向分量mgcos θ；摩擦反对相对运动趋势。 考试不会总以熟悉的标题出现：它可能被包装成图像、表格、坐标、单位、参数或现实情境。先还原“已知量—未知量—约束关系”，再决定使用哪条规则。先判断可能运动方向，再放摩擦；极限摩擦F=μR仅在limiting equilibrium。

Detailed English explanation. Choose and state a positive direction, isolate the body or system, and write equations with signs that match the diagram. Formulae such as SUVAT, F=ma and conservation laws are valid only when their modelling assumptions are satisfied.

变式推演 | Variation and transfer

若把例题中的具体数换成字母，应先建立一般关系再代值；若改变范围、方向、图形性质或随机条件，则必须重新检查原方法的适用条件。把答案遮住，尝试只凭题干写出第一行模型；第一行正确，后续计算才有方向。

第三道示范：审题、计算、复核 | Worked reasoning example

重新审视本课基础题：2 kg块在30° 光滑斜面上的加速度。

先把文字条件转换成数学关系，再依据本课方法推进。完整结果是：沿面力=2g sin30°，所以a=g sin30° =4.9 m/s。 复核时从结论逆向代回，并检查量级、正负、单位和定义域。若不一致，优先检查括号、指数、角度模式、方向约定或概率事件范围。

Cambridge 题型如何包装 | How the idea is examined

选择题常把典型失误设计成干扰项：漏掉第二个解、混淆百分数基数、把面积比当长度比、把独立当互斥或遗漏连续性修正。结构题常先要求 show/find 一个中间结果，再用 hence
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英文作答骨架 | Useful response frames

“From the given information, …” → “Using …, we obtain …” → “Substituting and simplifying gives …” → “Therefore the required value is …” . For interpretation, write “This means that … in the context of the question.” For validation, write “The value lies in the stated domain and satisfies the original condition.”

第三个迁移任务 | Transfer task

Why is friction not always μR?

独立完成后对照：F ≤ μR; equality holds only when motion is impending or limiting.

Exam check. Have you answered the command word, shown the decisive line of working, kept suitable accuracy, included units or direction, and considered every valid solution?

第 5 课 | 连接粒子与滑轮 | Connected particles

Lesson 5 of 8 · Detailed bilingual learning unit

概念拆解 | Concept and meaning

轻不可伸长绳使连接粒子加速度大小相同；光滑轻滑轮两侧张力相同。

English core explanation. Choose and state a positive direction, isolate the body or system, and write equations with signs that match the diagram. Formulae such as SUVAT, F=ma and conservation laws are valid only when their modelling assumptions are satisfied.

为什么方法成立 | Reasoning behind the method

分别为每个粒子列F=ma，再联立；不要把整个系统内部张力重复计算。

把知识点拆成可执行步骤 | A reliable method

1. Read the command word and identify the target quantity. 先圈出 find, show, prove, sketch, hence 或 explain。
2. Translate the information into equations, a labelled diagram, a table or a distribution.
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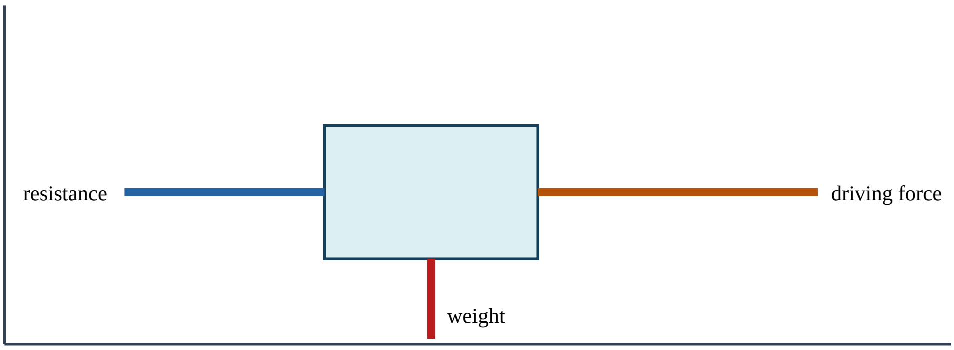

例题一：从题意到完整解答 | Worked example 1

2 kg和3 kg悬挂，求加速度(g=9.8)。

净驱动力=(32)g=9.8；总质量5；a=1.96 m/s。

Reading the mathematics. The first line identifies the structure; the middle lines show the transformation or calculation; the final line answers the question in context. Do not hide essential reasoning inside the calculator.

第 5 课（续）｜连接粒子与滑轮｜Connected particles

例题二：Cambridge 风格应用｜Cambridge-style application

Find the tension in the same system.

For 2 kg rising: T19.6=2(1.96), so T=23.52 N.

评分点如何产生｜Where marks come from

多数结构题把分数分给不同层次：建立正确模型或公式通常得到 method mark；正确代入并推进得到后续 method/accuracy marks；准确答案、单位、范围或结论得到 final accuracy mark。若前一步算错但方法仍一致，后续常可获得 follow-through，因此必须写出过程。证明题不能只写计算器结果，show that 题还要到达题目指定的精确形式。

English mathematical communication

Use complete mathematical sentences: “Substituting … gives …”, “Therefore …”, “Since the discriminant is zero …”, “The events are independent because …”, or “The negative sign shows motion opposite to the chosen positive direction.” When a question says hence, use the previous result. When it says exact, do not replace π, e, logarithms or radicals by rounded decimals.

易错辨析｜Common errors and repairs

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中途过早取整：后续答案超出允许误差。修正：保留精确值或计算器完整储存值。
只给答案：丢失方法分。修正：至少展示关键公式、代入与一次等价变形。
符号或单位遗漏：尤其是向量方向、平方/立方单位、概率区间和角度模式。
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> Detailed English explanation. Choose and state a positive direction, isolate the body or system, and write equations with signs that match the diagram. Formulae such as SUVAT, F=ma and conservation laws are valid only when their modelling assumptions are satisfied.

变式推演 | Variation and transfer

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第三道示范: 审题、计算、复核 | Worked reasoning example

重新审视本课基础题: 2 kg和3 kg悬挂, 求加速度(g=9.8)。

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Cambridge 题型如何包装 | How the idea is examined

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第三个迁移任务 | Transfer task

Find the tension in the same system.

独立完成后对照: For 2 kg rising: T19.6=2(1.96), so T=23.52 N.

> Exam check. Have you answered the command word, shown the decisive line of working, kept suitable accuracy, included units or direction, and considered every valid solution?

第 6 课 | 动量与冲量 | Momentum and impulse

Lesson 6 of 8 · Detailed bilingual learning unit

概念拆解 | Concept and meaning

动量mv有方向；封闭系统碰撞前后总动量守恒；冲量等于动量变化。

English core explanation. Choose and state a positive direction, isolate the body or system, and write equations with signs that match the diagram. Formulae such as SUVAT, F=ma and conservation laws are valid only when their modelling assumptions are satisfied.

为什么方法成立 | Reasoning behind the method

统一正方向，恢复系数不属于现行M1必需时不要乱加。

把知识点拆成可执行步骤 | A reliable method

1. Read the command word and identify the target quantity. 先圈出 find, show, prove, sketch, hence 或 explain。
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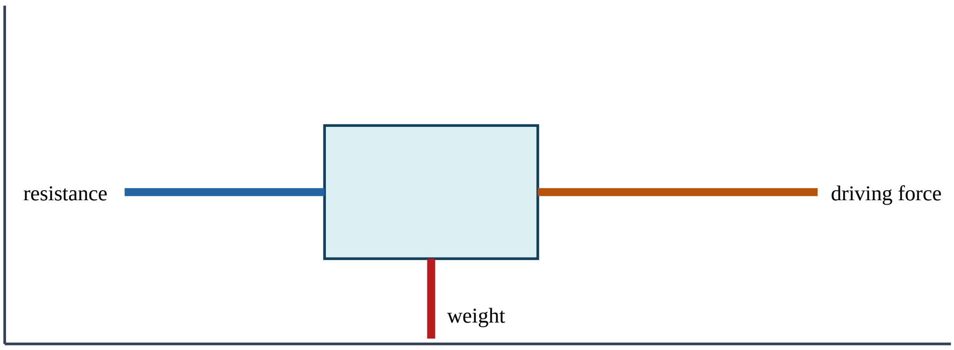

例题一：从题意到完整解答 | Worked example 1

2 kg以6 m/s右行，与1 kg静止物粘在一起，求共同速度。

初动量12；总质量3；v=4 m/s向右。

Reading the mathematics. The first line identifies the structure; the middle lines show the transformation or calculation; the final line answers the question in context. Do not hide essential reasoning inside the calculator.

第 6 课（续）丨动量与冲量丨Momentum and impulse

例题二：Cambridge 风格应用丨Cambridge-style application

A force of 5 N acts for 0.8 s. Find impulse.

Impulse=Ft=4 N s.

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多数结构题把分数分给不同层次：建立正确模型或公式通常得到 method mark；正确代入并推进得到后续 method/accuracy marks；准确答案、单位、范围或结论得到 final accuracy mark。若前一步算错但方法仍一致，后续常可获得 follow-through，因此必须写出过程。证明题不能只写计算器结果，show that 题还要到达题目指定的精确形式。

English mathematical communication

Use complete mathematical sentences: “Substituting … gives …”，“Therefore …”，“Since the discriminant is zero …”，“The events are independent because …”, or “The negative sign shows motion opposite to the chosen positive direction.” When a question says hence, use the previous result. When it says exact, do not replace π, e, logarithms or radicals by rounded decimals.

易错辨析丨Common errors and repairs

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> Detailed English explanation. Choose and state a positive direction, isolate the body or system, and write equations with signs that match the diagram. Formulae such as SUVAT, F=ma and conservation laws are valid only when their modelling assumptions are satisfied.

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第三道示范：审题、计算、复核｜Worked reasoning example

重新审视本课基础题：2 kg以6 m/s右行，与1 kg静止物粘在一起，求共同速度。

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第三个迁移任务｜Transfer task

A force of 5 N acts for 0.8 s. Find impulse.

独立完成后对照：Impulse=Ft=4 N s.

> Exam check. Have you answered the command word, shown the decisive line of working, kept suitable accuracy, included units or direction, and considered every valid solution?

第 7 课 | 功、能、功率 | Work, energy and power

Lesson 7 of 8 · Detailed bilingual learning unit

概念拆解 | Concept and meaning

功是力沿位移方向的分量×位移；动能mv，重力势能mgh，功率是做功率。

English core explanation. Choose and state a positive direction, isolate the body or system, and write equations with signs that match the diagram. Formulae such as SUVAT, F=ma and conservation laws are valid only when their modelling assumptions are satisfied.

为什么方法成立 | Reasoning behind the method

无非保守力时用机械能守恒；有阻力则把阻力做功纳入。

把知识点拆成可执行步骤 | A reliable method

1. Read the command word and identify the target quantity. 先圈出 find, show, prove, sketch, hence 或 explain。
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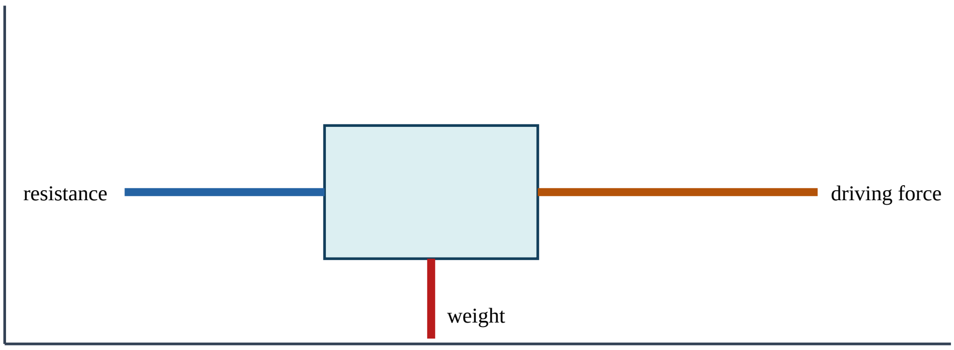

例题一：从题意到完整解答 | Worked example 1

3 kg物体从高5 m静止下落，忽略阻力，求落地速度。

mgh=0.5mv，v=√(2gh)=√98≈9.90 m/s。

Reading the mathematics. The first line identifies the structure; the middle lines show the transformation or calculation; the final line answers the question in context. Do not hide essential reasoning inside the calculator.

第 7 课（续）丨功、能、功率丨Work, energy and power

例题二：Cambridge 风格应用丨Cambridge-style application

A motor does 24 kJ in 15 s. Find power.

$P=W/t=24\ 000/15=1600$ W.

评分点如何产生丨Where marks come from

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> Detailed English explanation. Choose and state a positive direction, isolate the body or system, and write equations with signs that match the diagram. Formulae such as SUVAT, F=ma and conservation laws are valid only when their modelling assumptions are satisfied.

变式推演 | Variation and transfer

若把例题中的具体数换成字母，应先建立一般关系再代值；若改变范围、方向、图形性质或随机条件，则必须重新检查原方法的适用条件。把答案遮住，尝试只凭题干写出第一行模型；第一行正确，后续计算才有方向。

第三道示范：审题、计算、复核 | Worked reasoning example

重新审视本课基础题：3 kg物体从高5 m静止下落，忽略阻力，求落地速度。

先把文字条件转换成数学关系，再依据本课方法推进。完整结果是：mgh=0.5mv， v=√(2gh)=√98 ≈ 9.90 m/s。复核时从结论逆向代回，并检查量级、正负、单位和定义域。若不一致，优先检查括号、指数、角度模式、方向约定或概率事件范围。

Cambridge 题型如何包装 | How the idea is examined

选择题常把典型失误设计成干扰项：漏掉第二个解、混淆百分数基数、把面积比当长度比、把独立当互斥或遗漏连续性修正。结构题常先要求 show/find 一个中间结果，再用 hence
推进。上一问的精确形式通常是下一问最简洁的入口。

英文作答骨架 | Useful response frames

“From the given information, …” → “Using …, we obtain …” → “Substituting and simplifying gives …” → “Therefore the required value is …” . For interpretation, write “This means that … in the context of the question.” For validation, write “The value lies in the stated domain and satisfies the original condition.”

第三个迁移任务 | Transfer task

A motor does 24 kJ in 15 s. Find power.

独立完成后对照：P=W/t=24 000/15=1600 W.

> Exam check. Have you answered the command word, shown the decisive line of working, kept suitable accuracy, included units or direction, and considered every valid solution?

第 8 课 | M1 综合建模 | Mechanics synthesis

Lesson 8 of 8 · Detailed bilingual learning unit

概念拆解 | Concept and meaning

力学高分来自模型假设、方向约定、单位与分段事件。

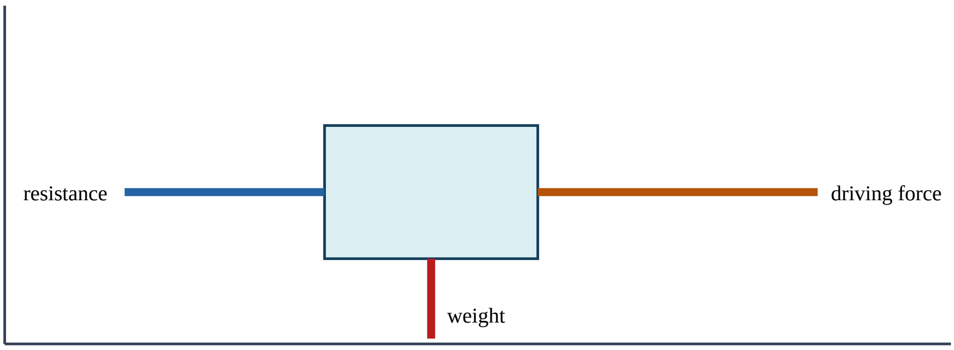
English core explanation. Translate between words, symbols, diagrams and numerical values. State assumptions, keep exact values during working, and test the final result against the original domain, scale and context.

为什么方法成立 | Reasoning behind the method

碰地、绳松、力改变都要新建阶段；最终检查速度符号和物理可行性。

把知识点拆成可执行步骤 | A reliable method

1. Read the command word and identify the target quantity. 先圈出 find, show, prove, sketch, hence 或 explain。
2. Translate the information into equations, a labelled diagram, a table or a distribution.
3. Choose a theorem or formula only after checking its assumptions.
4. Keep exact values through intermediate steps; round once at the end.
5. Check sign, size, unit, domain and whether all solutions have been found.



例题一：从题意到完整解答 | Worked example 1

为什么解得负速度不一定错？

它表示实际方向与选定正方向相反；应在结论中说明方向。

Reading the mathematics. The first line identifies the structure; the middle lines show the transformation or calculation; the final line answers the question in context. Do not hide essential reasoning inside the calculator.

第 8 课（续）｜M1 综合建模｜Mechanics synthesis

例题二：Cambridge 风格应用｜Cambridge-style application

State two modelling assumptions for a particle.

Its dimensions are negligible and rotational effects are ignored.

评分点如何产生｜Where marks come from

多数结构题把分数分给不同层次：建立正确模型或公式通常得到 method mark；正确代入并推进得到后续 method/accuracy marks；准确答案、单位、范围或结论得到 final accuracy mark。若前一步算错但方法仍一致，后续常可获得 follow-through，因此必须写出过程。证明题不能只写计算器结果，show that 题还要到达题目指定的精确形式。

English mathematical communication

Use complete mathematical sentences: “Substituting … gives …”, “Therefore …”, “Since the discriminant is zero …”, “The events are independent because …”, or “The negative sign shows motion opposite to the chosen positive direction.” When a question says hence, use the previous result. When it says exact, do not replace π, e, logarithms or radicals by rounded decimals.

易错辨析｜Common errors and repairs

条件未检查：把只适用于直角三角形、恒加速度、独立事件或正态模型的公式随意套用。修正：公式前写出适用结构。
中途过早取整：后续答案超出允许误差。修正：保留精确值或计算器完整储存值。
只给答案：丢失方法分。修正：至少展示关键公式、代入与一次等价变形。
符号或单位遗漏：尤其是向量方向、平方/立方单位、概率区间和角度模式。
漏解或增根：解三角、模、平方与对数方程后，回到原定义域逐一检查。

再推一步｜Extension and connection

把本课与相邻知识连接：同一对象通常可从符号、图像、数值表和现实情境四种表示来理解。若代数式得到的符号与图像趋势不一致，应回查运算；若概率或测量答案超出合理范围，应先检查模型而不是盲信计算器。这样的交叉检查是从“会算”走向“会解题”的关键。

> Exam-ready summary. Define or identify → model → calculate with visible steps → interpret → check. Your final statement should answer the exact question asked, with appropriate accuracy and units.

知识边界与相邻考点 | Scope and connections

力学高分来自模型假设、方向约定、单位与分段事件。 考试不会总以熟悉的标题出现：它可能被包装成图像、表格、坐标、单位、参数或现实情境。先还原“已知量—未知量—约束关系”，再决定使用哪条规则。碰地、绳松、力改变都要新建阶段；最终检查速度符号和物理可行性。

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